Hill Estimation of Left-Tail Risk for Intel (INTC)

Extreme Value Analysis in the AI Infrastructure Boom

Quantitative Risk Project

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Abstract

This paper examines whether a normal distribution provides an accurate picture of downside risk in Intel Corporation (INTC) stock. This question is especially important because Intel's role in the artificial-intelligence boom depends on an ambitious turnaround in foundry manufacturing and data-center chips. These opportunities may support future growth, but they also expose the stock to sharp changes in investor expectations.

Using 2,511 daily log returns from 2016-08-17 to 2026-08-14, the paper applies the Hill estimator from extreme value theory (EVT) to Intel's largest observed losses. The estimated left-tail index is $\hat{\alpha} = 2.202$ at $k = 167$, indicating that extreme losses occur more often than a Gaussian model would predict. A constant Gaussian 99% Value-at-Risk (VaR) is breached 37 times, compared with 25.1 expected breaches. These findings suggest that relying only on standard volatility or Gaussian VaR may understate Intel's exposure to unusually large down days.

1 Introduction and Motivation

The rapid growth of artificial intelligence has increased demand not only for graphics processing units, but also for the factories, server processors, and supporting infrastructure needed to operate AI systems. Intel participates in this expansion differently from a company such as Nvidia. Rather than leading the market for AI accelerators, Intel's opportunity comes mainly from manufacturing chips for other companies and supplying CPUs for data centers. Its place in the AI boom is therefore closely connected to a broader turnaround story.

That distinction matters for investors. Intel could benefit if customers want an alternative to TSMC or if data-center demand remains strong, but the company must also prove that it can manufacture advanced chips on schedule and attract profitable outside customers. As a result, Intel's stock can react sharply to a small number of important announcements. Good news about customers, production yields, or demand can support the rally, while a delay or guidance miss can quickly reverse expectations. This combination makes Intel a useful case study for downside tail risk.

One part of the investment case is Intel's potential to become an alternative to TSMC. Advanced-node capacity is limited, so customers may want a second source of manufacturing. The project outline reports that Intel Foundry revenue reached \$5.8 billion, up 31% year over year, while reported customer interest included Google, Tesla, and Microsoft. However, the same outline notes a \$2.1 billion operating loss and only \$293 million in external foundry revenue. The foundry business therefore creates both opportunity and risk: expectations may rise before outside demand and profitability are fully established.

In contrast, the more established source of AI-related earnings is Intel's Data Center and AI division. According to the project outline, the division generated \$2.5 billion in operating income

while revenue increased 59% year over year. This performance gives the turnaround a real operating foundation, but it does not remove the execution risk in foundry or the competition from AMD and TSMC.

These competing strengths and risks make tail-risk measurement especially relevant. The project outline describes a roughly 365% rise in Intel shares over the preceding year and several large moves around partnership news. After a rally of that size, historical standard deviation alone may not answer the most important risk question. An investor also needs to know whether unusually severe losses occur more often than a normal distribution would suggest. Possible triggers include manufacturing yield problems, a customer withdrawing, weak guidance, or geopolitical and export-control shocks. These events are uncommon, but their impact may be large.

2 Research Question and Approach

This paper asks a focused question: *Does a Gaussian model understate the frequency of extreme daily losses in Intel stock?* To answer it, the analysis uses two related tests. First, the Hill estimator [1] measures how quickly the probability of a large loss declines as the loss becomes more severe. Its key output is the tail index α; a smaller value means a heavier tail. Second, a Gaussian 99% VaR is compared with the losses that actually occurred. If the model is appropriate, approximately 1% of returns should fall below that threshold.

The two methods address the same issue from different directions. The Hill estimate describes the shape of the most extreme part of the return distribution, while the VaR backtest asks whether a familiar normal model produces the number of exceptions it promises. This connection allows the paper to move from a statistical result to a practical question about position size, risk limits, and model choice.

3 Data

Daily closing prices for INTC were taken from a Nasdaq-style historical export (CSV derived from Excel: `HistoricalData_1786949137896.csv`). The sample covers almost ten years, which provides enough observations to study rare losses while also introducing an important limitation: Intel in 2016 was not operating under the same turnaround and AI narrative as Intel in 2026. We compute log returns and retain all trading days so that the analysis uses a consistent daily series.

Table 1: Sample summary.

Field	Value
Ticker	INTC (Intel)
Date range	2016-08-17 – 2026-08-14
Calendar span	9.99 years
Daily prices	2,512
Daily log returns	2,511
Positive loss days	1,228

4 Methodology

4.1 Log returns and losses

Let P_t denote the closing price on day t. We begin by converting price levels into daily log returns,

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right). \tag{1}$$

Because the paper focuses on downside risk, these returns are then converted into nonnegative loss magnitudes,

$$L_t = \max(0,\, -r_t). \tag{2}$$

Thus, positive returns are assigned a loss of zero and negative returns become positive loss values. The Hill calculation uses only the strictly positive losses $\{L_t : L_t > 0\}$, giving 1,228 down days.

4.2 Hill estimator

After constructing the loss series, the observations are sorted from largest to smallest. Let $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(n)}$ denote these ordered loss magnitudes. For $1 \leq k < n$, the Hill estimator of the tail index is

$$\hat{\alpha}_k = \left[\frac{1}{k}\sum_{i=1}^{k} \ln\frac{X_{(i)}}{X_{(k+1)}}\right]^{-1}. \tag{3}$$

Under standard regular variation conditions, $\hat{\alpha}_k$ is consistent for the tail exponent governing

$$\Pr(X > x) \sim c\, x^{-\alpha}, \qquad x \to \infty. \tag{4}$$

In practical terms, Equation 4 says that the probability of a loss larger than x falls at a power-law rate. This decline is much slower than the decline in Gaussian tail probabilities, so the value of α helps show how strongly a normal model may understate extreme risk.

4.3 Threshold selection

The main judgment in a Hill analysis is the choice of k, the number of largest losses treated as tail observations. If k is too small, a few days control the estimate and make it unstable. If k is too large, ordinary losses enter the sample and bias the estimate away from the true tail. We therefore compute $\hat{\alpha}_k$ for $k = 1, \ldots, k_{\max}$ and select $k^\star$ from the most stable, approximately flat region of the Hill plot. The code formalizes this choice by minimizing the coefficient of variation over rolling windows with $k \geq k_{\min}$.

4.4 Gaussian Value-at-Risk backtest

Let $\hat{\mu}$ and $\hat{\sigma}$ be the sample mean and standard deviation of $\{r_t\}$. These parameters provide a simple benchmark against the EVT result. A constant one-day Gaussian VaR at confidence level λ is

$$\mathrm{VaR}_\lambda = \hat{\mu} + \hat{\sigma}\,\Phi^{-1}(1-\lambda), \tag{5}$$

where Φ is the standard normal CDF. For $\lambda = 0.99$, a breach occurs on day t if $r_t < \mathrm{VaR}_{0.99}$. If the Gaussian model and its constant parameters describe the data well, the expected breach rate is $1 - \lambda = 1\%$. This is a deliberately simple benchmark rather than a production trading model, but its failures are easy to interpret.

Moment existence. If tail behavior is Pareto-like with index α, moments of order p exist only when $p < \alpha$. Thus $\alpha > 2$ implies finite variance, while $\alpha > 4$ is required for finite kurtosis.

5 Empirical Results

5.1 Hill estimate

Table 2 summarizes the tail and VaR diagnostics. At $k^\star = 167$, we estimate $\hat{\alpha} = 2.2016$ with tail threshold $X_{(k^\star+1)} = 3.45\%$. This means that the model treats the 167 largest losses—those above approximately 3.45%—as observations from the extreme tail. Because $\hat{\alpha} \approx 2.2$, a literal Pareto extrapolation implies finite variance ($\alpha > 2$) but no finite fourth moment ($\alpha < 4$). The estimate should not be read as proof that Intel's true kurtosis is infinite; instead, it is evidence that the observed tail is much heavier than a Gaussian tail.

Table 2: Tail index and Gaussian VaR backtest.

Quantity	Estimate
Chosen threshold $k^\star$	167
Hill tail index $\hat{\alpha}$	2.2016
Threshold loss $X_{(k^\star+1)}$	3.45%
Gaussian 99% VaR (Eq. 5)	−6.61%
Observed breaches	37
Expected breaches (1% rule)	25.1
Breaches per year (observed)	3.70
Breaches per year (expected)	2.51

5.2 Visual diagnostics

Figure 1 displays the Hill plot; the selected $k^\star$ lies in a stable plateau. Figure 2 compares the empirical survival function to a power-law fit anchored at the threshold. Figure 3 contrasts the return histogram with a Gaussian density, highlighting excess tail mass. Figure 4 marks Gaussian 99% VaR breaches through time, including clustered extreme days. Taken together, the figures tell a consistent story rather than four separate ones: the return histogram first shows that normality is questionable, the Hill and log–log plots characterize the heavy tail, and the VaR chart shows the consequence in risk-model performance.

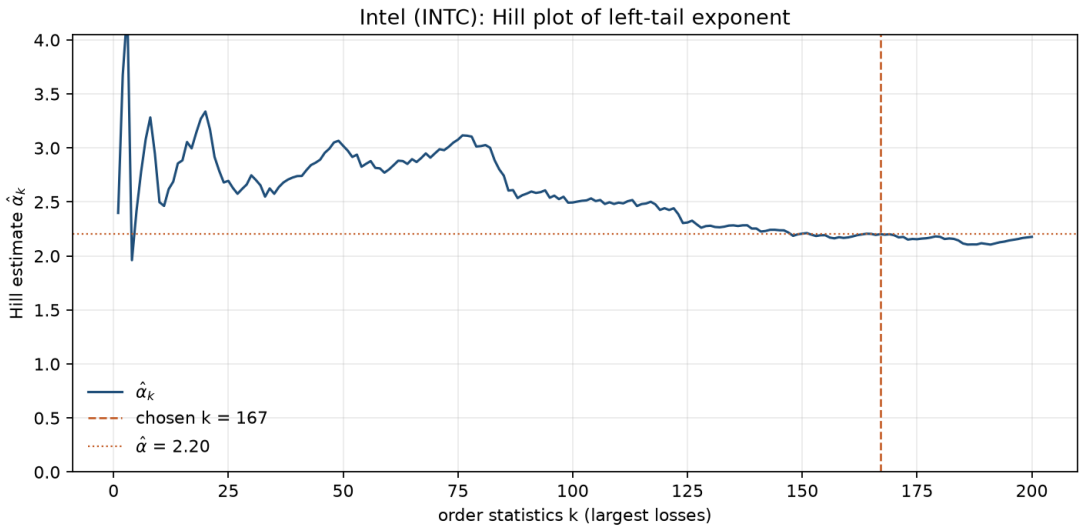


Figure 1: Hill plot $\hat{\alpha}_k$ versus order statistic k. The dashed line marks $k^\star = 167$; the dotted line marks $\hat{\alpha} = 2.20$.

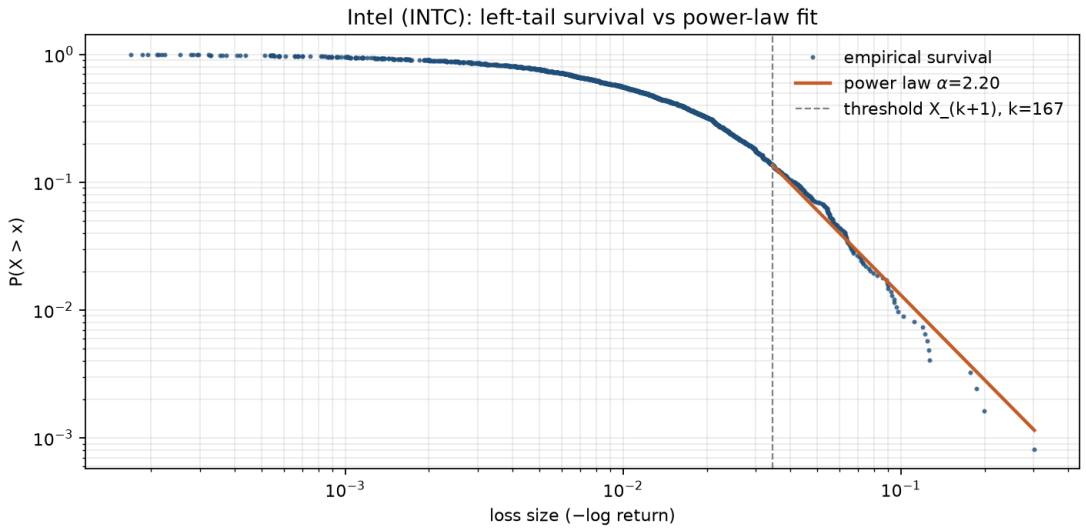


Figure 2: Log–log plot of empirical tail survival $\widehat{\Pr}(X > x)$ versus loss size, with power-law fit ($\alpha = 2.20$) above the threshold.

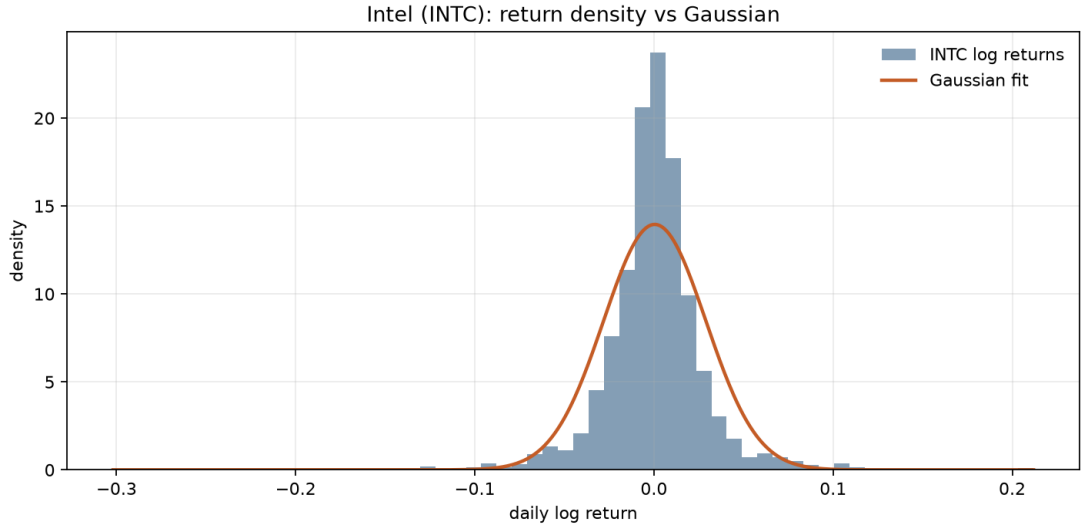


Figure 3: Daily log-return density versus a Gaussian fit, illustrating leptokurtosis.

6 What the Results Reveal

The clearest result is that the Gaussian model is too optimistic about Intel's downside tail. Across 2,511 returns, a 1% exception rate implies 25.1 breaches. The sample contains 37, or roughly 47% more than expected. In annual terms, the model anticipates about 2.51 breaches, while the data contain 3.70. The difference is economically meaningful because a VaR model is used precisely to describe rare losses; matching average volatility is not enough if the model misses the events that matter most.

The Hill estimate explains why the breach count is high. With $\hat{\alpha} \approx 2.2$, the loss tail decays slowly, meaning that moving farther into the tail does not reduce probability as quickly as it would under a normal distribution. This does not predict the date or cause of the next large decline. It does show that severe down days are a recurring feature of the historical return

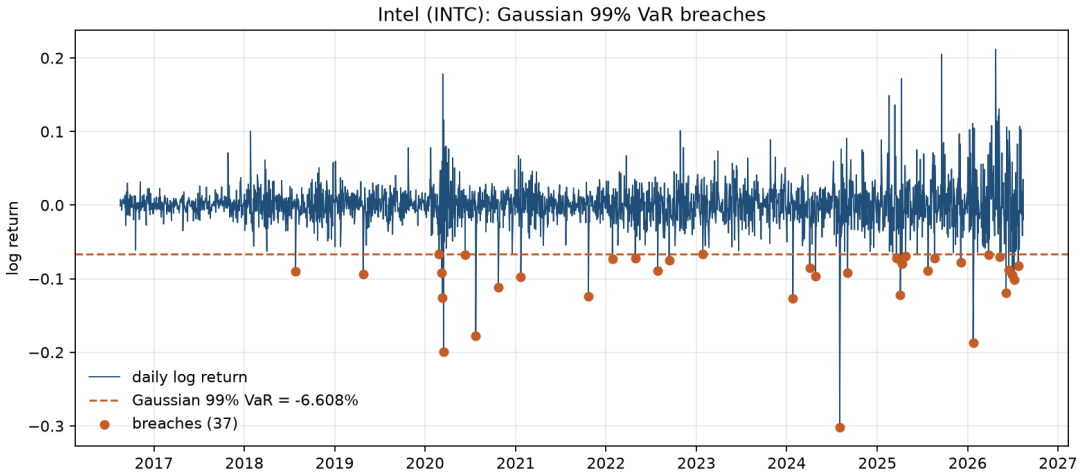


Figure 4: Daily log returns with constant Gaussian 99% VaR (Eq. 5). Orange markers indicate breach days ($N = 37$).

process rather than statistical impossibilities.

The result also connects back to Intel's business position. The AI and foundry strategy creates upside if Intel wins customers and executes successfully, but it concentrates expectations around a limited set of milestones. When those expectations change, the price adjustment can occur in a single session. The statistical finding therefore supports the economic argument in Section 1: Intel's downside risk is shaped not only by routine day-to-day volatility, but also by infrequent, high-impact information.

7 Decision Implications

The estimates do not produce a buy or sell recommendation on their own. They do, however, change how an investor, portfolio manager, or risk analyst should frame several decisions.

Position sizing. A position sized only from standard deviation may be too large because standard deviation compresses the entire distribution into one number. The high breach count and low tail index support using a smaller position, a larger loss budget, or a tail-aware risk measure when the goal is to keep a single INTC shock from dominating the portfolio.

Risk limits and capital buffers. Using the Gaussian 99% VaR of -6.61% as a hard worst-case boundary would be inappropriate; 37 observations crossed it. For internal risk limits, the threshold should be treated as a warning level rather than a maximum possible daily loss. A manager could supplement it with historical stress days, EVT-based VaR or Expected Shortfall, and scenario losses tied to foundry delays, customer cancellations, or weak guidance.

Diversification and hedging. Because Intel's largest moves may be connected to company-specific milestones, diversification across unrelated return drivers can reduce exposure more reliably than owning several semiconductor names with similar AI expectations. An investor who must hold the stock could also consider defined-risk hedges around earnings or major manufacturing announcements. The analysis does not price such a hedge, but it provides a reason to evaluate one.

Model selection and monitoring. For reporting ordinary volatility, a Gaussian model may still be a useful baseline. For capital allocation or stress testing, the evidence favors a model

that explicitly allows heavy tails. A practical next step would be to estimate rolling values of $\hat{\alpha}$ and compare pre-turnaround and AI-boom periods. If the rolling estimate falls, risk limits could be tightened; if it rises consistently across multiple thresholds and windows, a manager would have stronger evidence that downside tail risk is easing.

Fundamental decisions. The analysis is most useful when paired with a catalyst checklist. Foundry profitability, external customer revenue, 18A/14A yields, DCAI demand, guidance, and export restrictions are possible sources of abrupt repricing. The Hill estimate tells the decision-maker to take those scenarios seriously, while fundamental research helps assign plausible causes and decide whether the expected return justifies the risk.

8 Limitations and Further Work

Several limits prevent the estimate from being treated as a permanent law of Intel's stock. First, Hill estimates are sensitive to k. Although $k^\star = 167$ lies in a stable region, another defensible threshold would change $\hat{\alpha}$. Confidence intervals and alternative threshold rules would make the conclusion more robust.

Second, the sample pools different business regimes. Intel before the recent turnaround and Intel during the AI boom may not follow the same return process. A rolling Hill estimate, structural-break test, or sub-period comparison would show whether the tail itself changed.

Third, the Gaussian VaR benchmark uses constant $(\hat{\mu}, \hat{\sigma})$ and ignores volatility clustering. A GARCH-filtered EVT model would provide a fairer comparison with a modern conditional risk model. Finally, the fundamental figures in the purpose outline should be verified against primary filings and company disclosures before publication.

9 Conclusion

This paper began with a practical concern: Intel's AI opportunity may create substantial upside, but the same turnaround narrative can expose investors to sudden changes in expectations. The data support that concern. The Hill estimate of $\hat{\alpha} = 2.202$ indicates a heavy left tail, and the Gaussian 99% VaR is breached 37 times rather than the 25.1 times expected.

The main conclusion is not that a crash is certain or that Intel should be avoided. It is that Gaussian volatility alone gives an incomplete picture of the risk. For decisions involving position size, capital buffers, diversification, hedging, or stress scenarios, a heavy-tail model provides information that the normal model leaves out. The result is most valuable when combined with fundamental monitoring of the milestones that can trigger abrupt repricing.

Disclaimer

This document summarizes a quantitative exercise and conversational project outline. It is **not** investment advice or a sell-side research report. Verify all market and fundamental figures against primary sources before making investment decisions.

References

[1] B. M. Hill, A simple general approach to inference about the tail of a distribution, *Annals of Statistics*, 3(5):1163–1174, 1975.

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[3] A. J. McNeil, R. Frey, and P. Embrechts, *Quantitative Risk Management: Concepts, Techniques and Tools*, Princeton University Press, 2015.